

Substitution in Isabelle/HOL

```
theory Subst
  imports Main
begin

lemma
  “ $((x::nat) + y) * (a + b) = x * a + y * a + x * b + y * b$ ”
proof-
  have 1: “ $(x + y) * (a + b) = (x + y) * a + (x + y) * b$ ”
    apply(subst distrib_left)
    ..
  also have 2: “ $\dots = x * a + y * a + (x + y) * b$ ”
    apply(subst distrib_right)
    ..
  also have 3: “ $\dots = x * a + y * a + (x * b + y * b)$ ”
    apply(subst distrib_right)
    ..
  also have 4: “ $\dots = x * a + y * a + x * b + y * b$ ”
    apply(subst add.assoc[symmetric])
    ..
  finally show ?thesis
  .
qed

end
```